INSTALLATION, WIRING AND OPERATOR GUIDE

Lathe Digital Readout

Raspberry Pi Pico 2 + Dell Ubuntu display

Wiring the X/Z scales and LJ12A3-4-Z/BX spindle pickup, flashing firmware, starting the DRO, and checking readings before use.

X / Z scales
Quadrature A/B

>

Signal interface
3.3 V logic

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Pico 2
Counts / RPM

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Dell / Ubuntu
DRO display

Electrical safety: Pico GPIO is not 5 V tolerant. The LJ12A3-4-Z/BX probe uses a 6-36 V supply; never connect its supply wire to GP8. Connect the NPN output directly only if verified as an open-collector/sinking output: the Pico's pull-up then holds GP8 at 3.3 V. Otherwise use a compatible interface.

System settings in this project

Function	Firmware/application assumption
Mill X/Y/Z scales	X GP2/3; Y GP6/7; Z GP4/5
Lathe X/Z scales	X GP9/10; Z GP11/12; independent from mill scales
Spindle pickup	GP8 falling-edge timing; 4 pulses per spindle revolution assumed
Linear conversion	25,400 decoded counts per inch by default; verify against actual travel
Host connection	USB serial; separate LX/LZ and MX/MY/MZ counts plus RPM

This is a measurement and calculation aid, not a motion controller or safety interlock. It does not stop or control the lathe. Continue using guards, the machine manual, and independent measuring tools.

Identify the sensor interface first

Power down before wiring. Stop the spindle, isolate machine power as appropriate, unplug the Pico, prevent accidental movement, and keep wiring clear of rotating parts, full axis travel, chips, coolant, and mains/VFD wiring.

- Every Pico GPIO input must stay between 0 V and 3.3 V. 5 V, 12 V, 24 V, or negative-going signals can damage the Pico.
- Scale inputs GP2-GP7 and GP9-GP12 have no firmware pull-up/down. The Ditron scales default to TTL but may be ordered as EIA-422; confirm the output before choosing its interface. Never connect differential A-/B- directly to GPIO.
- GP8 has an internal pull-up to 3.3 V and detects falling edges. It is not a 5 V level shifter.
- The selected LJ12A3-4-Z/BX is a three-wire NPN normally-open inductive sensor rated 6-36 V; the listing gives brown = supply positive, blue = 0 V, black = output. Power it from a separate 12 V rail, not the Pico.
- For direct GP8 use, confirm the black lead is an open-collector/sinking output with no pull-up to the sensor supply. Connect blue/0 V to Pico GND for this non-isolated interface; the GPIO then sees only the Pico's internal 3.3 V pull-up. If output topology is unclear, add a suitable interface and verify its Pico-side level.
- Do not identify wires by color alone on other sensors. For non-isolated interfaces, connect a common signal reference only as specified. Do not assume the lathe frame or cable shield is signal ground.

LJ12A3-4-Z/BX tach pickup

1. Power the probe from a regulated 6-36 V supply. The listed PC-PSU 12 V rail is within range; fuse and insulate the supply wiring.
2. Mount a secure iron target at the probe's 4 mm nominal sensing distance. Do not use a loose target near rotating machinery.
3. With the probe disconnected from GP8, verify the black output switches low when metal is detected and is not pulled up to 5/12 V. Confirm the open-collector/sinking output from the sensor datasheet or by a safe measurement.
4. For a verified sinking output, connect black to GP8 and blue to Pico GND; leave GP8's internal 3.3 V pull-up enabled. Never connect brown or sensor supply voltage to a Pico pin.
5. Firmware assumes four falling edges per revolution. Use four targets or change the firmware pulse count to match your target arrangement. Validate RPM against a trusted tachometer with guards in place.

Safe first power-up: Flash firmware with all sensor signals disconnected. Verify the USB link, then add one correctly interfaced signal at a time with power removed between wiring changes.

Signal wiring plan

The table shows separate input pairs for both machines, plus each Pico 2 header pin. Both machines may remain wired to the Pico at the same time. Use the GP labels and official Pico 2 pinout to confirm board orientation before wiring.

> **Independent machine inputs:** The lathe and mill each have dedicated X and Z GPIO pairs; they do not share scale signal pins. Connect each conditioned scale pair to its labeled inputs. The firmware counts both machine profiles continuously, and the machine-selection screen chooses which readings to display. Power off before wiring changes.

Machine / signal	Pico GPIO	Physical pin	Use
Mill X scale A / B	**GP2 / GP3**	4 / 5	Mill X quadrature input
Mill Z scale A / B	**GP4 / GP5**	6 / 7	Mill Z quadrature input
Mill Y scale A / B	**GP6 / GP7**	9 / 10	Mill Y quadrature input
Spindle sensor black output	**GP8**	11	Falling-edge tach input
Lathe X scale A / B	**GP9 / GP10**	12 / 14	Lathe X quadrature input
Lathe Z scale A / B	**GP11 / GP12**	15 / 16	Lathe Z quadrature input
Sensor blue / signal 0 V	**GND**	3 or 8	Common reference for this non-isolated NPN output

```
Mill X interface OUT-A/B (safe 0-3.3 V) -----> Pico pins 4/5   (GP2/GP3)
Mill Z interface OUT-A/B ---------------------> Pico pins 6/7   (GP4/GP5)
Mill Y interface OUT-A/B ---------------------> Pico pins 9/10  (GP6/GP7)
Sensor black OUT (verified NPN sinking) ----> Pico pin 11 (GP8)
Lathe X interface OUT-A/B (safe 0-3.3 V) ----> Pico pins 12/14 (GP9/GP10)
Lathe Z interface OUT-A/B -------------------> Pico pins 15/16 (GP11/GP12)
Sensor blue 0 V ----------------------------> Pico GND pin 3 or 8
Sensor brown +6-36 V -----------------------> Separate 12 V rail; never GPIO

Scale output interface/supply: follow the scale supplier's documentation.
```

Signal-level connection map only; it is not a power-supply schematic. Fit the correct differential receiver, isolator, pull-up, or level shifter based on actual device documentation.

Assembly and routing

- Use insulated connectors and strain relief suitable for machine vibration, chips, and coolant.
- Route each signal pair with its return or as the interface specifies. Separate low-level signals from VFD, motor, contactor, and mains wiring.
- Secure cables and check full carriage/cross-slide travel with power isolated so nothing pulls, pinches, or fouls.
- If an axis direction is reversed, power down and swap that axis's A and B logic outputs at the interface, then retest.
- Verify one falling tach edge per intended target. Do not add arbitrary capacitors or pull-ups; they may distort pulses or conflict with the interface.

Flash and start the Pico 2

Flash the supplied UF2 on Windows

1. Stop the DRO on the Dell. Unplug the Pico from the Dell and use a USB data cable on Windows.
2. Hold BOOTSEL while plugging the Pico into the PC. Release when detected. Device Manager may show **RP2350 Boot**; the boot volume commonly appears as **RPI-RP2**.
3. Copy `build-pico2/lathe_dro_pico.uf2` from the project folder onto the Pico boot volume. Do not format or initialize it.
4. Wait for the volume to disappear as the Pico restarts. Reconnect it to the Dell normally, without BOOTSEL.

Flash the supplied UF2 on Linux

1. Stop the DRO and disconnect the Pico from the Dell. With a USB data cable, hold BOOTSEL while plugging the Pico into the Linux PC, then release it. The boot drive is commonly named **RPI-RP2** and may mount automatically.
2. Open the drive in the file manager. If it is not shown, identify the Pico boot partition with `lsblk -o NAME,LABEL,MOUNTPOINTS`, then mount/open that partition in the file manager. Do not format or repartition it.
3. In a terminal opened in the project folder, copy the UF2 to the drive's actual mount path (commonly `/media/$USER/RPI-RP2`):

   ```
   cp build-pico2/lathe_dro_pico.uf2 /media/$USER/RPI-RP2/
   sync
   ```

   If mounted at `/mnt/pico`, use `sudo cp build-pico2/lathe_dro_pico.uf2 /mnt/pico/` instead, then run `sync`.
4. Wait for the boot drive to disappear as the Pico restarts. Eject/unmount if it remains mounted, then reconnect the Pico to the Dell normally, without BOOTSEL.

Optional firmware rebuild on Ubuntu

The project uses CMake, Ninja, ARM GCC, and Pico SDK 2.3.1.

```
sudo apt update
sudo apt install cmake ninja-build gcc-arm-none-eabi git python3
git clone --branch 2.3.1 --recurse-submodules \
  https://github.com/raspberrypi/pico-sdk.git pico-sdk
git -C pico-sdk submodule update --init --recursive
chmod +x build_pico2.sh
./build_pico2.sh
```

Output: `build-pico2/lathe_dro_pico.uf2`. Building does not flash the Pico; use BOOTSEL.

Set up the Dell Ubuntu display

A graphical desktop/session is required for Pygame; headless SSH alone cannot display the app.

```
sudo apt install python3 python3-venv python3-pip
cd /path/to/lathe-dro-project
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
sudo usermod -aG dialout "$USER"
# Log out/in after changing group membership
chmod +x run_dro.sh
./run_dro.sh
```

The app auto-detects a Pico USB serial port. If necessary, specify the actual port, e.g. `./run_dro.sh --port /dev/ttyACM0`. To stop a foreground DRO, press `Ctrl+C` in its terminal. Restart after reconnecting/reflashing.

Optional demo: `./run_dro.sh --demo --windowed`. Simulated readings are not for calibration or machining.

> **Serial format:** current firmware sends lines like `LX:123,LZ:-456,MX:10,MY:20,MZ:30,RPM:300` at about 60 updates/second. The lathe readout uses LX/LZ; the mill readout uses MX/MY/MZ. Older firmware packets remain accepted but cannot provide separate scale counts. Raw counts restart at zero after Pico reboot; establish the work reference again.

Coordinate macros and geometry helpers

From the mill readout, select **MACROS**. These helpers display coordinates and calculations only; they do not command axis movement, spindle operation, or tool compensation.

Tool	Inputs and result
PCD / bolt circle	Enter center X/Y, pitch-circle diameter, whole hole count (2–24), and start angle. The screen lists numbered absolute X/Y hole coordinates, evenly spaced counterclockwise from the start angle.
Linear array	Enter start X/Y, hole count (2–24), pitch, and angle. The result lists each hole coordinate along the angled line.
Grid	Enter start X/Y, rows, columns, X/Y pitch, and rotation angle. Up to 24 hole coordinates are listed row by row.
Midpoint / distance	Enter two DRO values to find their midpoint or absolute shoulder-to-shoulder distance.
Cutter radius compensation	Enter the nominal straight-edge point, cutter diameter, travel angle, and LEFT/RIGHT side. The output is the offset tool-center point. This calculation covers a straight edge only; calculate/verify corner joins separately.
Calculator	STANDARD provides add, subtract, multiply, and divide. TRIG provides sin, cos, tan and inverse functions; input angles and inverse outputs are in degrees. SLOPE reports rise/run, percent, and angle.
Sub-datums	Choose one of eight slots, save the live mill X/Y/Z counts as a named datum, then select it to use those counts as the absolute DRO origin. Saved slots can also be deleted.

Coordinates and persistence: Pattern coordinates are absolute in the currently selected mill work datum; set the center/start values in that coordinate system. Sub-datums are saved in `mill_sub_datums.json`. Pico encoder counts restart at zero after a Pico reboot, so re-establish the machine reference and recapture any required datums before relying on them.

Verify before machining: Check each listed coordinate and compensation side independently, use safe clearances, and confirm the tool, workholding, and setup. These calculators are not a CNC controller or safety interlock.

Verify before machining

Lathe X and Z scale calibration

The app defaults to **25,400 decoded counts per inch**. Scale vendors may define cycles differently; SETUP calibration measures an effective scale factor but does not prove the scale's stated accuracy.

1. Stop the spindle. Select INCH/MM on the main screen before entering calibration distance.
2. Measure a long known straight-axis travel with a trusted standard. Approach the start consistently and account for backlash.
3. In **SETUP**, select the X or Z distance field, enter the known travel, position at the start, and click **CAPTURE X START** or **CAPTURE Z START**.
4. Move that axis by the entered distance, then click **CALIBRATE X** or **CALIBRATE Z**. The factor uses raw count change, independent of zero and display modes.
5. For X, enter actual cross-slide (radial) movement, not twice that amount from the DIA display. Recheck over a second distance and in both directions when safe.
6. Use the +/- 0.0001 buttons for fine adjustment only after checking mounting, missed counts, backlash, and signal integrity.

> **Persistence:** calibrated X/Z factors and fine adjustments are saved in `scale_calibration.json` beside the app. Mill factors are separate in `mill_scale_calibration.json`. Factors correct the base 25,400-counts/in assumption; verify against a trusted standard before machining.

Readout controls

Control	Meaning
ZERO X / ZERO Z	Set origin for the selected axis and mode.
ABS / INC	Absolute or incremental display; entering INC starts a fresh incremental origin.
RAD / DIA	X radius or diameter; diameter display is twice radial travel.
INCH / MM	Display units and calibration-entry units.
T-1 ... T-9	Select tool ID. Offsets apply in ABS when both T-1 and selected tool have offsets.
TOOL SET	Capture T-1 as master. For T-2 through T-9, capture both axes or enter a signed X offset relative to T-1 with the on-screen keypad. Entry follows current units and RAD/DIA mode; a manual X edit preserves the tool's saved Z offset. Stored in `tool_offsets.json`.
FEED / SPEED / TAPER / COORD	Reference calculators only; they do not operate the lathe. Verify chart and setup values independently.
SETUP	Calibrate X/Z from known travel, fine-adjust factors, and change display resolution. Factors persist in `scale_calibration.json`.
POWER	Requests Ubuntu shutdown after confirmation; does not stop the lathe.

Boring estimate

Enter the measured starting-hole diameter, position the tool at the starting bore wall, and capture X. The app adds twice the calibrated radial X travel. Choose X direction so the estimate increases while boring outward. Verify final diameter with a bore gauge.

RPM verification

Firmware measures falling-edge intervals and currently assumes four pulses/revolution. With the LJ12A3-4-Z/BX NPN NO sensor, use four secure iron targets at equal spacing or change the firmware pulse count. Confirm the sensor's switching rate is adequate at intended spindle speed and compare against a trusted tachometer with guards in place. RPM is advisory and does not control the spindle.

Commissioning checks

Symptom	Checks
Pico not found	Reconnect in normal mode; use a data USB cable; check Linux serial permissions/device path; restart DRO after flashing.
Lathe scale does not count	Check conditioned X A/B at GP9/10 or Z A/B at GP11/12.
Mill scale does not count	Check conditioned X A/B at GP2/3, Y A/B at GP6/7, or Z A/B at GP4/5.
Axis direction reversed	Power off and swap that axis's A/B interface outputs; retest known travel.
Erratic counts	Check receiver type, voltage, return path, loose terminals, electrical noise/cable routing, shield termination, and scale alignment. Do not guess pull-ups/capacitors.
RPM stays zero	Check the sensor's 6-36 V supply, iron target/gap, verified NPN sinking output, GP8 wiring, and secure target mounting.
RPM wrong by ratio	Count falling edges per revolution; firmware assumes four. Ensure the probe can switch fast enough.
Readings drift	Compare several distances; check calibration, missed counts, backlash, scale mounting, and mechanical alignment.

Before first use

[ ] Exact scale and LJ12A3-4-Z/BX sensor part numbers/output specifications recorded.
[ ] Required scale interface installed; tach output confirmed NPN sinking/open-collector.
[ ] All Pico GPIO signals verified between 0 and 3.3 V.
[ ] Sensor supply/ground/signal wiring checked; no sensor supply voltage reaches a GPIO.
[ ] Wiring insulated, secured, labeled, and clear through full travel.
[ ] X/Z direction and travel checked against trusted standards in multiple positions.
[ ] RPM checked against a trusted tachometer with guards in place.
[ ] Operator confirms ABS/INC, RAD/DIA, unit, and tool selection before measuring.

Stop using the DRO if readings jump, reverse unexpectedly, disagree with trusted measurements, or wiring is damaged. Isolate power and correct the cause before machining. This system is not a safety interlock or motion controller.

Project-specific assumptions match this repository's current firmware/application. Verify the installed scale output and LJ12A3-4-Z/BX sensor wiring against supplier documentation and confirm all Pico GPIO levels remain safe.